

Yiyang Lu

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EDUCATION

IIIS, Tsinghua University

Undergraduate Study

Sep 2024 - Now

IIIS, Tsinghua University

Pre-College Program

Sep 2023 - Jun 2024

RESEARCH INTERESTS

Computer Vision, Generative Models, Reinforcement Learning

EXPERIENCES

Undergraduate Research in Computer Vision

Supervised by Professor Kaiming He, Massachusetts Institute of Technology

Feb 2025 - Now

- Conduct research on generative models, including Diffusion models, Flow Matching, and Normalizing Flows.
- Develop and deploy models on Google Cloud Platform using TPU and JAX.

Undergraduate Research in Robotics

Supervised by Professor Huazhe Xu, Tsinghua University

Sep 2024 - Jun 2025

- Gain experiences on robotics and computer vision, especially designing diffusion models and vision encoder for imitation learning.
- Implementing ideas and training models on real robots and simulators, such as Metaworld, Maniskill, Robotwin.

Internship in Computer Vision

Tencent

Jul 2024, Jul 2025

- Worked on multi-modal AI, especially extracting features from images and texts.
- Selected as "Challenging Star" in the internship program.

PUBLICATIONS

Improved Mean Flows: On the Challenges of Fastforward Generative Models

Zhengyang Geng, Yiyang Lu*, Zongze Wu, Eli Shechtman, J. Zico Kolter, Kaiming He*

<https://arxiv.org/abs/2512.02012>

H³DP: Triply-Hierarchical Diffusion Policy for Visuomotor Learning

Yiyang Lu, Yufeng Tian*, Zhecheng Yuan*, Xianbang Wang, Pu Hua, Zhengrong Xue, Huazhe Xu*

<https://arxiv.org/abs/2505.07819>

AWARDS

National Scholarship

China

2025

Tsinghua Academy Talent Development Program Scholarship

Tsinghua University

2024

Freshman Scholarship (Second Prize)

Tsinghua University

2024

Gold Medal (Top 10)

39th Chinese Physics Olympiad

2022

PROJECTS

Flow-made Anime Characters

2025.5

https://github.com/PeppaKing8/dl_project

- Project for the course *Deep Learning* at IIIS, Tsinghua University.
- This project is selected as *the most popular project (top 1)*.

Realistic Multi-Material Interaction Simulation Pipeline

2024.12

<https://github.com/Lyy-iiis/ACG-Project>

- Project for the course *Advanced Computer Graphics* at IIIS, Tsinghua University.
- This project is selected for class *oral presentation (top 3)* and *the most popular project (top 3)*.

Do LLMs Outperform Multi-task Learning Expert in Medical Report Generation?

2024.12

<https://github.com/Lyy-iiis/Crayon-new>

- Project for the course *Machine Learning* at IIIS, Tsinghua University.
- This project is selected for class *oral presentation (top 3)*.

Randomized Techniques in Graph Algorithms

2024.10

<https://github.com/PeppaKing8/algdesign-project>

- Project for the course *Algorithm Design* at IIIS, Tsinghua University.
- This project receives *6 bonus points* for creativity.

Music Image Transfer

2024.6

https://github.com/Lyy-iiis/LLM_project

- Project for the course *Introduction to Large Language Model Application* at IIIS, Tsinghua University.
- The project is further developed by *ZhiPu AI*.

SKILLS

Programming Languages: Python, C/C++, Bash, Assembly, Scala

Tools: PyTorch, JAX, MATLAB, Git, LaTeX, Docker, GDB, SQL

Languages: English (Fluent), Chinese (Native)

ADDITIONAL INFORMATION

Transcript at IIIS, Tsinghua University

All of my **professional courses** taken at IIIS, Tsinghua University is shown below.

Year-Semester	Course Title	Credit	Grade
2023-Autumn	Introduction to Computer Science	3	A+
	Introduction to Programming in C/C++	2	A+
	Calculus A (1)	5	A
	Linear Algebra	4	A+
	General Physics (2)	4	A+
2024-Spring	Introduction to Computer Systems	4	A+
	Introduction to Large Language Model Application	2	A
	Mathematics for Computer Science and Artificial Intelligence	4	A+
	Calculus A (2)	5	A+
	General Physics (1)	4	A
	Quantum Computer Science	4	A+
2024-Autumn	Advanced Computer Graphics	3	A+
	Natural Language Processing	3	A+
	Machine Learning	4	A+
	Algorithm Design	4	A+
	Artificial Intelligence: Principles and Techniques	3	A+
2025-Spring	Computer Vision	3	A
	Deep Learning	3	A
	Multi-modal Machine Learning	3	A
2025-Summer	Type-safe Modern System Practice	2	A

Self-Studied/Audited courses

- Abstract Algebra, IIIS, Tsinghua University
- The Missing Semester of Your CS Education, MIT, <https://missing.csail.mit.edu/>
- Introduction to Algorithms, MIT, <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-spring-2020/>
- Design and Analysis of Algorithms, MIT, <https://ocw.mit.edu/courses/6-046j-design-and-analysis-of-algorithms-spring-2015/>
- CS224n: Natural Language Processing with Deep Learning, Stanford, <http://web.stanford.edu/class/cs224n/>
- CS231n: Deep Learning for Computer Vision, Stanford, <http://cs231n.stanford.edu/>
- Intro to Computer Systems, CMU, <https://www.cs.cmu.edu/afs/cs/academic/class/15213-f15/www/schedule.html>
- CS285: Deep Reinforcement Learning, UC Berkeley, <https://rail.eecs.berkeley.edu/deeprlcourse>